Sustainable Transportation Coordinator

Carpooling & bike-sharing optimization for campus. Route matching & environmental impact tracking.

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Abstract

The Sustainable Transportation Coordinator project addresses mobility challenges at Universidad Distrital, Engineering campus, where parking shortage, transportation inefficiencies, and environmental concerns create the need for more sustainable commuting alternatives. This study presents a systems analysis focused on carpooling and bike-sharing coordination, using thorough direct observation, physical measurements, and process documentation to identify transportation habits, user perceptions, main barriers to adoption, and key operational requirements for a future mobility support system integrated with campus sustainability and security needs. The expected outcome of this first-stage analysis is to identify stakeholder needs, critical system elements, and improvement opportunities that can support the later design of a transportation coordination platform aimed at reducing congestion, improving commuting efficiency, and promoting lower-emission travel options.

Introduction

Sustainable transportation systems on university campuses represent a critical intervention for addressing chronic mobility challenges, including parking shortages, traffic congestion, and

greenhouse gas emissions from private vehicle dependency. Research by González and Ferreira highlights that traditional campus parking paradigms assume "el estacionamiento debe ser abundante, gratuito y basado en los requisitos mínimos de estacionamiento," frequently leading to overprovision exceeding 200% of actual demand and perpetuating environmental inefficiencies. Similarly, studies in Latin American contexts confirm that "cars dominate the campus vehicle fleet at both entrances (70% and 82%)" with carpooling interventions capable of reducing CO and NOx emissions by 16% and CO2 by 26%. These challenges are particularly pronounced within engineering faculties across multiple urban campuses, where high student and staff volumes exacerbate infrastructure limitations amid dense metropolitan traffic patterns. This systems analysis investigates the Sustainable Transportation Coordinator—a platform integrating carpooling optimization, bike-sharing coordination, route matching algorithms, and environmental impact monitoring with campus security and sustainability systems. Through primary data collection across three engineering faculty locations, this study analyzes mobility patterns, infrastructure constraints, and user coordination dynamics in order to establish the basis for a future sustainable transportation system.

System Overview and Scope

The university operates in a dense urban environment affected by traffic congestion and a high carbon footprint, where circulation on nearby public roads is frequently obstructed by vehicles associated with institutional activity. In response to this context, the proposed system is a smart mobility management solution designed to coordinate the flow of vehicles and bicycles within the institution. Its main purpose is to optimize the use of physical infrastructure in order to address carbon emissions, campus mobility constraints, and the availability of parking spaces for

students and professors. The system is also expected to generate indirect benefits for surrounding residents by improving public space and reducing the obstruction of nearby roads caused by institutional transport demand.

The system promotes lower-emission mobility alternatives, especially bicycle use, while also improving the efficiency of available parking through shared-use strategies such as carpooling and bike-sharing. This approach seeks to reduce underutilization during low-demand periods and excessive pressure during peak times, when parking occupancy becomes disproportionate and inefficient. In the current context, a large share of institutional space is allocated to cars, but average occupancy is not optimal, since car spaces often remain unused while motorcycle users experience shortages. To improve this situation, the system must analyze variables such as class schedules, user demand, and vehicle capacity in order to maximize space utilization. In the case of bike-sharing, the system must also manage real-time bicycle availability, docking stations, and preventive maintenance needs.

System Boundaries: The analysis is restricted to the **Calle 40 campus (Sede 40)**, the surrounding **Calle 34** areas, and the **ECCI** university location.

Main Components: Common transit routes, parking infrastructure (basements and street-level), bicycle/scooter zones, and the university's digital registration app.

Stakeholders: Students, Professors.

Data Collection Plan and Execution

Following the workshop requirements, three primary data collection methods were implemented:

1. Physical Measurements & Direct Observation:

- **Calle 40 Basement:** Dedicated to motorcycles with a flexible, non-fixed quota system.
- **Calle 40 Street:** Approximately 90 motorcycles park along the external road daily.
- **Bicycle/Scooter Zone:** Estimated capacity for 50 units.
- **Calle 34 Zone:**Limited space for approximately 15 bicycles, 0 cars and motorcycles only on the platform.
- **ECCI Campus:** Zero internal parking availability; users rely on paid parking lots of the ecci university.

2. Structured Surveys (Habits & Perception):

A structured survey was conducted with members of the engineering faculty community in order to identify mobility habits, dominant transport modes, parking-related difficulties, and willingness to use shared mobility alternatives such as carpooling and bike-sharing. The survey included questions on user role, frequency of campus visits, main transport mode, arrival time, travel time, parking availability, shared travel practices, and interest in institutional coordination tools. This instrument complemented direct observation, physical measurements, and process documentation by incorporating user-centered evidence on behavioral patterns and perceived mobility constraints.

3. Process Documentation:

- **Access Protocol:** Users must register their vehicle in the university app (no cost). Physical entry is controlled manually by security staff based on real-time visual capacity.
- **Coordination:** Peer-to-peer organization currently occurs through informal WhatsApp and Facebook groups, primarily for safety and shared costs rather than parking availability.

Primary Data Presentation

The following table summarizes the parking infrastructure conditions observed across the three engineering faculty locations during peak academic hours. Data were collected through direct observation and physical measurements conducted on-site at each facility

Location	**Vehicle Type**	**Estimated Capacity**	**Observations**
Calle 40 (Basement)	Motorcycles	Flexible	Managed by security staff.
Calle 40 (External road)	Motorcycles	90	High impact on public space.
Calle 40 (Zone)	Bikes/Scooters	50	No active loan system exists.
Calle 34	Mixed(Bikes/ Motorcycles)	7 - 10	High demand relative to available space, frequently at or near capacity.

ECCI	All Vehicles types	0	No internal parking. Users must pay external paid facilities

Table 1. Parking infrastructure capacity and observed conditions across engineering faculty locations.

Process documentation further identified that vehicle access is regulated through a university mobile application, where users must register their vehicle prior to entry at no cost. However, physical access is ultimately controlled by security personnel based on real-time visual capacity assessment, which operates independently of the digital registration data. Informal peer-to-peer coordination for shared mobility and inter-campus travel occurs primarily through WhatsApp and Facebook groups, driven by cost and safety motivations rather than parking availability management.

Survey Results Summary

Survey Variable	Main option	Result
User profile	Student	98%
Dominant Transport mode	Public transport mode	60.8%
Most common arrival time	Before 8:00	54.9%
Willingness to use carpooling	Willing	34,3%
Willingness to use bike-sharing	Yes	74.5%

Table 2. Summary of structured survey results from engineering faculty users.

Survey Results analysis

Once the surveys are done, some behaviors are observed such as that the must of people use public transportation, that is to say that although the issue of motorcycles and bicycles affects many people, it is not the majority of the university population, it is a part of it, also that the majority of people are at the university from eight in the morning, that is to say that from eight until approximately two in the afternoon is where it usually happens that more parking spaces and spaces are needed, in addition to this the results show a greater acceptance of the proposal of bicycles than that of shared cars, thus being something that perhaps we could give more emphasis as a solution to these problems.

Data Quality Assessment

The collected data provide a useful initial characterization of campus mobility conditions; however, several limitations must be acknowledged. First, some measurements were based on direct field estimation rather than continuous automated monitoring, which may reduce precision in capacity and occupancy values. Second, the observations reflect specific locations and time periods, so the results may not capture all temporal variations across different days, weather conditions, or academic schedules. Third, access dynamics depend partly on manual decisions by security staff, which introduces operational variability that is not fully controlled by the registration system itself. In addition, some mobility practices, such as informal ride-sharing coordination through digital groups, were identified through process documentation and contextual interpretation, which may limit the consistency of user-level behavioral evidence. Despite these constraints, the combination of direct observation, physical measurements, and

process documentation provides sufficiently reliable evidence to identify bottlenecks, recurring patterns, and system-level improvement opportunities for the first stage of analysis

System Analysis Report

Elements and Relationship Analysis

The system exhibits high sensitivity to environmental and institutional variables:

- **Weather -> Choice:** Precipitation events trigger a shift from bicycles/motorcycles to public transport or grouped ride-sharing.
- **Security -> Flow:** This involves the security staff, their decision-making determines the "Full/Empty" state of the system, often independent of the app's registration data.

Sensitivity Analysis

- **Peak Time Variability:** The system reaches critical saturation at **12:00 PM**. During this window, even a small increase in demand (e.g., 5 additional motorcycles) causes a "spillover" effect, forcing illegal parking on Calle 40.
- **Cost Sensitivity:** The lack of parking at ECCI creates a "shared-cost" incentive, making students at this location the most likely early adopters for a formalized carpooling platform.

Emergent Behavior

- **Self-Organizing Inter-campus Transit:** In the absence of official university shuttles, students have evolved a manual carpooling system (grouping for Ubers) to minimize costs and transit time.

- **Entrance Bottlenecks:** Manual verification at the basement entrance creates a queue that obstructs the main street, affecting the local community and urban mobility.

Critical Assessment

- The system is limited by scarce parking capacity, manual access control, and the lack of internal parking at ECCI, which together create inefficiencies and recurring bottlenecks.

Visual System Representation

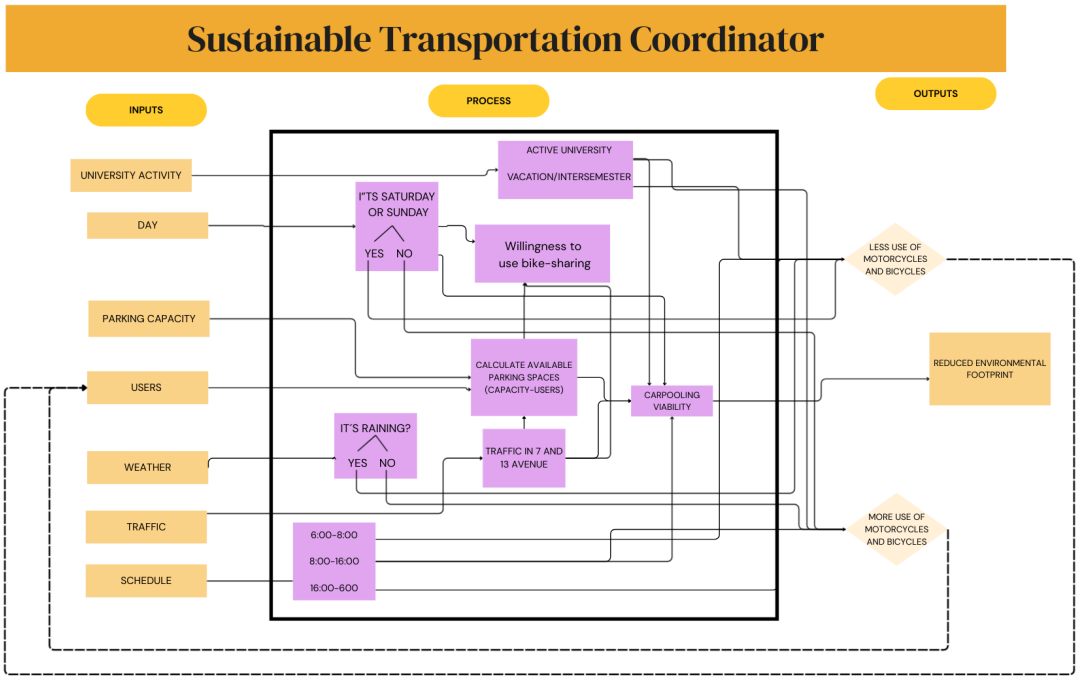

Diagram1. Parking infrastructure capacity and observed conditions across engineering faculty locations.

Conclusion

Systemic Imbalance and Critical Infrastructure

The analysis confirms that while physical infrastructure is limited, a "proto-carpooling" culture already exists through informal digital groups and shared trips, though it lacks official and proactive institutional support. A critical bottleneck is identified involving the Calle 34 and ECCI

locations, whereas the Calle 40 facility shows greater flexibility for system implementation. The total absence of parking at Calle 34 and ECCI forces students to rely entirely on Calle 40 for safety and space reasons. This reveals a systemic imbalance between infrastructure supply and institutional demand, highlighting a failure in providing accessible mobility solutions for the entire engineering community.

Operational Disconnection and Data Integrity

There is a significant operational inconsistency between the university's vehicle registration system and the manual entry control performed by security personnel. This "decoupling" implies that the system lacks real-time data integrity, as physical access is granted based on visual assessment rather than the digital database. This disconnect results in unpredictable availability patterns and inefficient space management.

Readiness for Change

Although private car usage is uncommon among students, there is a well-established informal culture of shared mobility via WhatsApp and Facebook to manage costs and safety. This indicates that the community has reached a level of socio-technical readiness for a formal Carpooling and Bike-sharing system that centralizes and optimizes these existing informal efforts.

Emergent Behaviors and Urban Externalities

The presence of motorcycles parked daily on the public roads outside Calle 40 confirms that the system boundaries extend beyond the campus walls. This externalization of the parking issue creates a negative emergent behavior that obstructs public space and affects local residents. Such

findings justify the urgent need for an internal optimization strategy to improve the institution's relationship with its urban operating environment.

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